CS3 HOOK DOCUMENT

DS 4002 – Fall 2025, Dean Paler
CASE STUDY REPOSITORY:
https://github.com/deanpaler/DS4002_CS3_CharlottesvilleCrimeTemperatureCorrelation

INTRODUCTION
Imagine this. You're the analyst called in by Charlottesville's city council during a sweltering summer. Crime reports seem to rise with the heat, and officials are pressing for answers. Is this just coincidence, or is there a deeper connection between temperature and public safety?They want evidence, and you're the one tasked with uncovering the truth.

In this case study, you'll step into the role of a data consultant investigating whether hotter days are truly linked to higher crime counts in Charlottesville. If the connection exists, your findings could shape how the city prepares for heat waves, allocates resources, and protects its residents, especially students navigating life in a college town.

CONTEXT
Across the world, researchers have discovered the patterns that as temperatures rise so do certain types of crime. In fact, scholars in criminology and environmental psychology have noted that heat has the potential to increase stress and aggression, which may lead to increased chances of conflict. Simultaneously, warmer weather gathers more people outdoors and into shared spaces, fueling more opportunities for both social interaction and potential crime. These signs suggest that higher temperatures may be correlated with increased rates of crime. Considering that past studies have emphasized that hotter temperatures can lead to aggression and violence, this raises questions about how communities should adapt in an increasingly warming world.

YOUR TASK
You have been provided with two datasets and a template Python notebook in the case study repository. The crime dataset includes five years of daily updates with details such as report time, offense description, reporting officer, and street/block. You will then retrieve current weather data using the Meteostat Python library, which includes weather station ID, timestamp, and metrics such as average/min/max temperature. With the guidance of the provided Python script, your mission is to analyze these datasets and uncover whether a statistically significant relationship exists between temperature and crime in Charlottesville.

Once you have completed your analysis based on the provided Python script, you will compile all your materials into your own GitHub repository and create a final slidedeck that communicates your findings. Have fun!